

Evaluation results (mIoU, OA & per-class IoU)

Parsed from `eval.log` files (and loveda `train.log`, which has no eval run) under `experiments/`. Re-run `scripts/extract_evaluation_tables.py` to refresh.

Dataset setup (resolution & split sizes)

All images are resized to **1008×1008** (config `dataset.image_size`) for training and evaluation.

Dataset	Resolution (crop)	Train	Val	Test
massachusetts_buildings	1008×1008	1,233	36	90
massachusetts_roads	1008×1008	9,972	126	441
potsdam	1008×1008	2,765	691	2,016
uavid	1008×1008	8,000	2,800	150
vaihingen	1008×1008	276	68	398
loveda	1008×1008	2,522	1,669	—
whu_building	1008×1008	5,732	1,228	1,228

Trainable parameters (by method)

Method	Trainable params	Total params	Trainable %
Linear Probing	2,299,652	456,339,460	0.50%
LoRA	4,539,139	460,877,828	0.98%
Adapter	337,891	456,676,580	0.07%

(From Massachusetts Buildings runs; slight variation with `num_classes`.)

Summary table (all runs)

Dataset	Method	mIoU	OA
loveda	Linear Probing	0.4237	0.5838
loveda	LoRA	0.5663	0.7411
loveda	Adapter	0.5518	0.7288
massachusetts_buildings	Linear Probing	0.7381	0.9080
massachusetts_buildings	LoRA	0.8278	0.9415
massachusetts_buildings	Adapter	0.8087	0.9335
massachusetts_roads	Linear Probing	0.7048	0.9638
massachusetts_roads	LoRA	0.8128	0.9798
massachusetts_roads	Adapter	0.8077	0.9790
potsdam	Linear Probing	0.7186	0.8170
potsdam	LoRA	0.8806	0.9306
potsdam	Adapter	0.8735	0.9293
uavid	Linear Probing	0.4554	0.7440
uavid	LoRA	0.6106	0.8697
uavid	Adapter	0.6619	0.8932
vaihingen	Linear Probing	0.6917	0.8186
vaihingen	LoRA	0.8326	0.9101
vaihingen	Adapter	0.8152	0.9048
whu_building	Linear Probing	0.8877	0.9682
whu_building	LoRA	0.9492	0.9862
whu_building	Adapter	0.9403	0.9837

Comparison by dataset (methods as columns)

Dataset	Linear Probing (mIoU)	Linear Probing (OA)	LoRA (mIoU)	LoRA (OA)	Adapter (mIoU)	Adapter (OA)
loveda	0.4237	0.5838	0.5663	0.7411	0.5518	0.7288
massachusetts_buildings	0.7381	0.9080	0.8278	0.9415	0.8087	0.9335
massachusetts_roads	0.7048	0.9638	0.8128	0.9798	0.8077	0.9790
potsdam	0.7186	0.8170	0.8806	0.9306	0.8735	0.9293
uavid	0.4554	0.7440	0.6106	0.8697	0.6619	0.8932
vaihingen	0.6917	0.8186	0.8326	0.9101	0.8152	0.9048
whu_building	0.8877	0.9682	0.9492	0.9862	0.9403	0.9837

Per-class IoU by dataset

loveda

Class	Linear Probing	LoRA	Adapter
background	0.3271	0.5533	0.5473
building	0.5934	0.6925	0.6617
road	0.5107	0.5874	0.5869
water	0.5649	0.7087	0.7054
barren	0.1676	0.3656	0.3472
forest	0.3282	0.4126	0.3984
agriculture	0.4737	0.6441	0.6158
mIoU	0.4237	0.5663	0.5518
OA	0.5838	0.7411	0.7288

massachusetts_buildings

Class	Linear Probing	LoRA	Adapter
background	0.8946	0.9308	0.9216
building	0.5817	0.7249	0.6958
mIoU	0.7381	0.8278	0.8087
OA	0.9080	0.9415	0.9335

massachusetts_roads

Class	Linear Probing	LoRA	Adapter
background	0.9627	0.9790	0.9782
road	0.4469	0.6465	0.6372
mIoU	0.7048	0.8128	0.8077
OA	0.9638	0.9798	0.9790

potsdam

Class	Linear Probing	LoRA	Adapter
impervious surface	0.7268	0.9070	0.9051
building	0.7778	0.9612	0.9567
low vegetation	0.6002	0.8078	0.8056
tree	0.6542	0.8089	0.7985
car	0.8338	0.9182	0.9016
mIoU	0.7186	0.8806	0.8735
OA	0.8170	0.9306	0.9293

uavid

Class	Linear Probing	LoRA	Adapter
building	0.6686	0.8962	0.9191
road	0.6596	0.8176	0.8391
tree	0.5989	0.7652	0.7915
low vegetation	0.4111	0.6231	0.6858
moving car	0.3233	0.5047	0.5230
static car	0.4027	0.5395	0.7027
human	0.1232	0.1276	0.1722
mIoU	0.4554	0.6106	0.6619
OA	0.7440	0.8697	0.8932

vaihingen

Class	Linear Probing	LoRA	Adapter
impervious surface	0.7215	0.8784	0.8721
building	0.8128	0.9322	0.9143
low vegetation	0.4987	0.7427	0.7323
tree	0.7262	0.8192	0.8157
car	0.6991	0.7905	0.7415
mIoU	0.6917	0.8326	0.8152
OA	0.8186	0.9101	0.9048

whu_building

Class	Linear Probing	LoRA	Adapter
background	0.9632	0.9838	0.9809
building	0.8123	0.9146	0.8997
mIoU	0.8877	0.9492	0.9403
OA	0.9682	0.9862	0.9837

Parsed log files

- experiments/sam3_adapter_loveda/train.log — loveda / Adapter: mIoU=0.5518, OA=0.7288
- experiments/sam3_linear_probing_loveda/train.log — loveda / Linear Probing: mIoU=0.4237, OA=0.5838
- experiments/sam3_lora_loveda/train.log — loveda / LoRA: mIoU=0.5663, OA=0.7411
- experiments/sam3_adapter_massachusetts_buildings/sam3_adapter_massachusetts_buildings_eval_2026-03-24_10-40-26/eval.log — massachusetts_buildings / Adapter: mIoU=0.8087, OA=0.9335
- experiments/sam3_linear_probing_massachusetts_buildings/sam3_linear_probing_massachusetts_buildings_eval_2026-03-09_06-55-50/eval.log — massachusetts_buildings / Linear Probing: mIoU=0.7381, OA=0.9080

- experiments/sam3_lora_massachusetts_buildings/
sam3_lora_massachusetts_buildings_eval_2026-03-07_21-54-09/eval.log —
massachusetts_buildings / LoRA: mIoU=0.8278, OA=0.9415
- experiments/sam3_adapter_massachusetts_roads/
sam3_adapter_massachusetts_roads_eval_2026-04-08_19-54-55/eval.log — massachusetts_roads /
Adapter: mIoU=0.8077, OA=0.9790
- experiments/sam3_linear_probing_massachusetts_roads/
sam3_linear_probing_massachusetts_roads_eval_2026-03-08_21-48-20/eval.log —
massachusetts_roads / Linear Probing: mIoU=0.7048, OA=0.9638
- experiments/sam3_lora_massachusetts_roads/
sam3_lora_massachusetts_roads_eval_2026-03-07_22-03-41/eval.log — massachusetts_roads /
LoRA: mIoU=0.8128, OA=0.9798
- experiments/sam3_adapter_potsdam/sam3_adapter_potsdam_eval_2026-03-16_23-38-43/eval.log —
potsdam / Adapter: mIoU=0.8735, OA=0.9293
- experiments/sam3_linear_probing_potsdam/
sam3_linear_probing_potsdam_eval_2026-02-25_22-40-38/eval.log — potsdam / Linear Probing:
mIoU=0.7186, OA=0.8170
- experiments/sam3_lora_potsdam/sam3_lora_potsdam_eval_2026-02-25_15-46-31/eval.log —
potsdam / LoRA: mIoU=0.8806, OA=0.9306
- experiments/sam3_adapter_uavid/eval_2026-03-24_10-55-55/eval.log — uavid / Adapter:
mIoU=0.6619, OA=0.8932
- experiments/sam3_linear_probing_uavid/sam3_linear_probing_eval_2026-03-15_17-02-00/
eval.log — uavid / Linear Probing: mIoU=0.4554, OA=0.7440
- experiments/sam3_lora_uavid/sam3_lora_uavid_eval_2026-03-14_23-00-44/eval.log — uavid /
LoRA: mIoU=0.6106, OA=0.8697
- experiments/sam3_adapter_vaihingen/sam3_adapter_vaihingen_eval_2026-03-16_23-34-44/
eval.log — vaihingen / Adapter: mIoU=0.8152, OA=0.9048
- experiments/sam3_linear_probing_vaihingen/
sam3_linear_probing_vaihingen_eval_2026-02-26_13-57-59/eval.log — vaihingen / Linear
Probing: mIoU=0.6917, OA=0.8186
- experiments/sam3_lora_vaihingen/sam3_lora_vaihingen_eval_2026-02-26_13-47-26/eval.log —
vaihingen / LoRA: mIoU=0.8326, OA=0.9101
- experiments/sam3_adapter_whu_building/eval_2026-03-24_11-07-39/eval.log — whu_building /
Adapter: mIoU=0.9403, OA=0.9837
- experiments/sam3_linear_probing_whu_building/eval_2026-03-12_19-16-13/eval.log —
whu_building / Linear Probing: mIoU=0.8877, OA=0.9682
- experiments/sam3_lora_whu_building/sam3_lora_whu_building_eval_2026-03-11_23-36-25/eval.log
— whu_building / LoRA: mIoU=0.9492, OA=0.9862